

Chang Hwan Kim

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Education

M.S., Mechanical Engineering, Yonsei University | Seoul, S. Korea

Exp. Aug. 2026

- Thesis (expected): Design and Model Predictive Control of a Dual Reluctance-Lorentz 6-DOF Magnetic Levitator
- Advisor: Prof. Jun Young Yoon
- GPA: 4.23/4.30 (= 3.97/4.00)

B.S., Mechanical Engineering, Yonsei University | Seoul, S. Korea

Mar. 2019 – Aug. 2024

- GPA: 4.13/4.30 (= 3.92/4.00), **Rank #1 out of 145** in Major (**Summa Cum Laude**)
- 1.5-year absence due to military service (Jan. 2021 – Jul. 2022)

Research Interests

Vehicle dynamics control, Model predictive control, Control barrier function, Mechatronics

Journal Publications & Manuscripts Under Review

1. **C.H. Kim**, H.M. Yoon, J.W. Jeong, E.K. Kim and J.Y. Yoon, “DNN-Approximated Offset-Free MPC on FPGA for 10-kHz Constrained In-Plane Control of 6-DOF Magnetic Levitation System”, IEEE Transactions on Industrial Informatics (under review)
2. **C.H. Kim**, H.M. Yoon, J.W. Jeong, E.K. Kim and J.Y. Yoon, “Design and Control of Dual Reluctance-Lorentz 6-DOF Magnetic Levitator for In-Plane Long-Stroke Motion and Low-Power Vertical Levitation”, IEEE/ASME Transactions on Mechatronics (minor revision)
3. H.M. Yoon, **C.H. Kim**, K.J. Kang, J.M. Sung, E.K. Kim ... and J.Y. Yoon, “Active 6-DOF Magnetic Levitator for Fully-Automated High-Throughput Large-Range Single-Mode Optical Fiber Alignment”, IEEE Transactions on Industrial Electronics (accepted)
4. S.M. Lee, Y.S. Kwon, J.W. Jeong, K.J. Kang, **C.H. Kim** and J.Y. Yoon, “Identification and Reduction of Normal-Direction Force Ripples in Permanent-Magnet Linear Synchronous Motors”, IEEE Transactions on Industrial Electronics [\[PDF\]](#)
5. H.M. Yoon, S.M. Lee, J.W. Jung, K. Lee, J.J. Jung, **C. H. Kim** and J.Y. Yoon, “Study on Characteristics of SPGD Algorithm in Coherent Beam Combining with Phase and Tip-tilt Control Speed Differences”, Journal of the Korean Society for Precision Engineering [\[PDF\]](#)

Conference Presentations

1. **Control Barrier Function-Based Safety-Critical Control of 6-DOF Magnetic Levitation System**

Korean Society for Precision Engineering Spring Conference

May. 2026

2. **Offset-Free Model Predictive Control of 6-DOF Magnetic Levitator for High-Precision Trajectory Tracking**

Korean Society for Precision Engineering Autumn Conference

Nov. 2025

3. **Gain-Scheduled Feedforward-Feedback Control for Reducing Lateral Acceleration and Jerk of Vehicle During Lane Changing Maneuver**

International Conference on Precision Engineering on Sustainable Manufacturing (PRESM2025)

Jul. 2025

4. **Control Method for Maintaining Dynamic Performance of a 6-DOF Magnetic Levitator During Payload Mass Changes**

International Conference on Precision Engineering on Sustainable Manufacturing (PRESM2024)

Jul. 2024

Honored with outstanding presentation award

Research Experiences & Selected Projects

Graduate Researcher, Yonsei University (Advisor: Prof. Jun Young Yoon)

Sep. 2024 – present

Undergraduate Researcher, Yonsei University (Advisor: Prof. Jun Young Yoon)

Jan. 2023 – Aug. 2024

1. **FPGA-Based, DNN-Approximated Offset-Free MPC for 6-DOF Magnetic Levitator** (ongoing project)

Mar. 2025 – present

- Designed an offset-free model predictive controller (MPC) with a disturbance observer.
- Currently implementing deep neural network-approximated MPC on FPGA, targeting > 10 kHz control.

2. **Design and Control of 6-DOF Magnetic Levitator for Long-Stroke Zero-Power Gravity Compensation**

Jan. 2024 – Jun. 2025

- Designed and fabricated a 6-DOF magnetic levitation system based on a novel electromagnetic working principle.
- Developed and implemented 6-DOF feedback and feedforward controllers for precision motion control.

3. **Vehicle Lateral Dynamics Control for Reducing Lateral Acceleration and Jerk During Lane Changes**

Sep. 2024 – Feb. 2025

- Developed a gain-scheduled preview feedforward controller and integrated it with a lane-keeping LQR feedback controller.

- Used a single-track vehicle model with Ackermann steering geometry and designed a Luenberger-PI observer for state estimation.
4. **Force Ripple Reduction Control of Permanent-Magnet Linear Synchronous Motor** Mar. 2023 – Mar. 2024
- Designed a controller that reduces both normal- and tangential-direction force ripples.
 - Proposed a motor parameter estimation method that induces force ripples in iron-cored PMLSM.
5. **Adaptive Control of Low-power 6-DOF Hybrid-reluctance-type Magnetic Levitator** Mar. 2023 – Dec. 2023
- Designed a control method that estimates the payload of the system using a disturbance observer.
 - Implemented a controller that maintains the dynamic performance of the system during payload changes.

University-Industry Projects

1. **Laser CBC SPGD Control Modeling & Simulation Development: in collaboration with *LIG Nex1 Co., Ltd.*** Apr. 2024 – Mar. 2025
- Designed a stochastic parallel gradient descent (SPGD) algorithm for laser coherent beam combining over 19 channels.
 - Developed a MATLAB simulator for laser beam modeling and SPGD algorithm performance evaluation.
2. **Independent Drive & Control System for Styler Moving Hanger: in collaboration with *LG Electronics Inc.*** Nov. 2024 – Oct. 2025
- Designed a novel drive mechanism enabling independent movement of the styler’s moving hanger.
 - Developed a motion controller that compensates for actuator cogging force and bearing friction.
3. **Intelligent Controller Development for PAPS-based robotic C-arm: in collaboration with *GENORAY Co., Ltd.*** Sep. 2025 – present
- Designing sub-optimal collision-free paths and trajectories for a 4-DOF robotic C-arm using an A* algorithm.
 - Implemented and tested these methods in a MATLAB simulation environment to ensure avoidance of patient-area intrusion.

Honors and Awards

- Graduated with Highest Honor (Summa Cum Laude, Top 1%), Yonsei University** Aug. 2024
- Given to only four students from the College of Engineering.
- Outstanding Presentation Award, PRESM2024** Aug. 2024
- Received at the PRESM2024 international conference.
- Semester Honors, Yonsei University** Spring 2019, Fall 2019, Fall 2023, Spring 2024

Scholarships

- Scholarship for the Accelerated Bachelor's-Master's Program, Yonsei University** Sep. 2024 – Aug. 2026
- Awarded for academic excellence to support early admission to the master’s program.
 - Full tuition coverage for 4 semesters.
- Brain Korea 21 Plus Scholarship, National Research Foundation (NRF) of Korea** Sep. 2024 – Aug. 2026
- Awarded to support graduate research in strategic fields under Korea’s national initiative.
- Daesang Foundation Scholarship, Daesang Group** Mar. 2020 – Aug. 2024
- Given to outstanding undergraduate students to support academic growth and contribute to society.
 - Full tuition coverage for 6 semesters.
- Yonsei Eminence Scholarship, Yonsei University** Sep. 2019 – Feb. 2020
- One-third tuition coverage for one semester for academic excellence.

Teaching Assistant Experiences

- MEU3014 - Mechatronics** Fall 2024
- Topics covered: electronics, sensors, motors, linear system control theory, and control practice using a microcontroller.
- MEU3007 - Independent study** Spring 2025
- Supervised an undergraduate student for their undergraduate thesis.

Patents & Copyrights

1. **Multi-Channel Laser Coherent Beam Combining Simulation Software, C-2025-018066**

Other Experiences

- Republic of Korea Army | Inje, S.Korea** Jan. 2021 – Jul. 2022
- Served as an infantry rifle squad leader in the Opposing Force regiment (2nd mountain regiment, KCTC).
- Korean Red Cross Blood Management Center | Seoul, S.Korea** Jun. 2017 – present
- Regular blood donor (20 donations since Jun 2017).

Skills

Programming & Software: MATLAB/Simulink, LabVIEW, Python, C, ANSYS Electronics, Autodesk Inventor, LaTeX

Tools: MATLAB MPC, Control System, Optimization, Deep Learning, Robotics Toolboxes; Vehicle Dynamics Blockset; MPT3

Languages: Korean (Native), English (Full professional proficiency)